

1. Software Requirements Specification (SRS)

1.1 Overall Description

CampusConnect acts as a modular web application structured around a micro-service or clean monolithic architecture, serving student buyers, sellers, service providers, event organizers, and student job seekers.

1.2 System Interfaces

- **User Interface:** Web Application optimized for Desktop, Tablet, and Mobile viewport sizes (Responsive PWA-Progressive Web App).
- **External APIs:**
 - SMTP / Email Service(e.g., SendGrid) for email verification and notification dispatch.
 - Cloud Storage API(e.g., AWS S3) for media and document assets(we can decide to choose a different platform).

2. Functional Requirements Document (FRD)

The functional requirements detail the behavior and specific business rules of the CampusConnect platform.

Requirement ID	Module	Description	Priority
<i>FR-AUTH-01</i>	User Auth	The system shall restrict registration to valid university email addresses (.edu.cm or institutional domain).	High

<i>FR-AUTH-02</i>	User Auth	The system shall send an automated verification link/code upon registration.	High
<i>FR-MKT-01</i>	Marketplace	Users shall be able to create, edit, soft-delete, and mark items as "Sold".	High
<i>FR-MKT-02</i>	Marketplace	Sellers shall be able to attach up to 5 high-resolution images per listing.	Medium
<i>FR-SRV-01</i>	Services	Service providers shall be able to post skill listings featuring hourly rates, skills, and portfolio links.	High
<i>FR-JOB-01</i>	Opportunities	Verified recruiters/administrators shall be able to post internships and jobs with application deadlines.	High
<i>FR-JOB-02</i>	Opportunities	Students shall be able to upload a PDF resume and apply directly through the platform listing.	High
<i>FR-EVT-01</i>	Campus Events	Event hosts shall be able to schedule events with location, date, time, category, and capacity flags.	Medium
<i>FR-MSG-01</i>	Messaging	The system shall enable direct, real-time messaging between prospective buyers/clients and sellers/providers.	High
<i>FR-SCH-01</i>	Search/Filter	Users shall be able to search globally or filter specifically by category, price range, condition, and posting date.	High

3. Non-Functional Requirements Document (NFRD)

3.1 Performance Requirements

- **NFR-PERF-01:** System pages must achieve an initial load time under **2.0 seconds** over standard 4G connectivity.
- **NFR-PERF-02:** Search results must return within **500 milliseconds** for a database containing up to 100,000 active listings.

3.2 Security Requirements

- **NFR-SEC-01:** All network traffic must be encrypted in transit using **TLS 1.3 / HTTPS**.
- **NFR-SEC-02:** User passwords must be hashed using a strong hashing function (e.g., **bcrypt** or **Argon2**) prior to storage.
- **NFR-SEC-03:** Media uploads must undergo automated MIME-type validation and malware scanning before storage.

3.3 Reliability & Availability

NFR-REL-01: System availability must maintain **99.5% uptime** during active academic semesters.

NFR-REL-02: Data must be backed up incrementally every 24 hours with disaster recovery point objective (RPO) ≤ 1 hour.

3.4 Usability & Accessibility

NFR-USA-01: User Interface must comply with **WCAG 2.1 Level AA** standards.

NFR-USA-02: Layout must be fully responsive across viewport widths ranging from 320px to 2560px.

4. User Stories

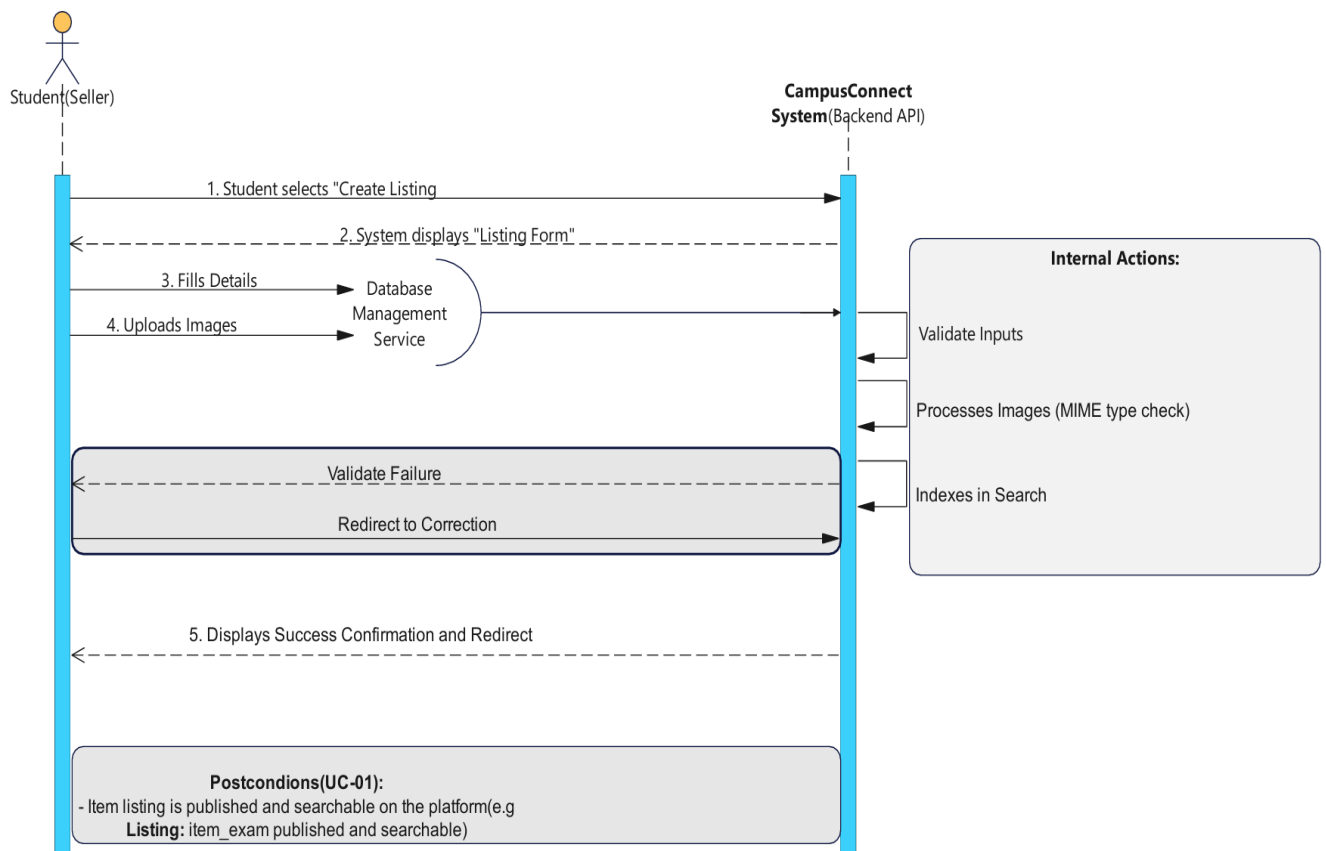
- **US-01 (Marketplace Buyer):** As a student buyer, I want to filter marketplace listings by category and maximum price so that I can find affordable textbooks within my budget.
- **US-02 (Marketplace Seller):** As a graduating student, I want to upload photos and descriptions of my used furniture so that I can sell it to incoming students before moving out.
- **US-03 (Service Provider):** As a student photographer, I want to create a service page showcasing my portfolio so that campus student organizations can hire me for events.

- **US-04 (Job Seeker):** As a senior student, I want to search for local tech internships and submit my resume directly on the platform so that I can secure employment before graduation.
- **US-05 (Event Attendee):** As a student, I want to view a timeline of upcoming campus events so that I can participate in extra-curricular activities.
- **US-06 (Inquirer/Buyer):** As a student interested in a listing, I want to send a direct message to the seller so that we can agree on a safe, physical meeting location on campus.

5. Detailed Use Case Descriptions

5.1 Use Case UC-01: Create Marketplace Listing

- **Primary Actor:** Verified Student (Seller)
- **Preconditions:** User is logged into an authenticated account.
- **Postconditions:** Item listing is published and searchable on the platform.



Main Success Scenario:

1. User navigates to the Marketplace module and clicks "**Create Listing**".
2. System presents the listing form (Title, Description, Price, Category, Condition, Upload Photos).
3. User fills out required fields, attaches image files, and clicks "**Publish**".
4. System validates inputs, processes image uploads, indexes the listing for search, and displays a success confirmation message.
5. System redirects the user to the published listing page.

Extension/Alternative Flows:

3a. Validation Failure: If required fields are missing or photo files exceed size constraints >5MB per image):

1. System highlights invalid fields with specific error messages.
2. User corrects errors and submits again.

3b. Unverified Email: If user email verification status is pending:

1. System blocks submission and prompts user to verify email first.

5.2 Use Case UC-02: Search & Contact Seller

- **Primary Actor:** Verified Student (Buyer)
- **Preconditions:** Buyer is logged into the application.
- **Postconditions:** A message thread between Buyer and Seller is initialized.

Main Success Scenario:

1. User enters keywords into the global search bar and selects the "**Marketplace**" filter.
2. System displays matching listing items sorted by relevance or posting recency.
3. User applies additional filters (e.g., Price Range: 1000 FCFA - 30,000 FCFA, Condition: "Like New").
4. System refreshes results instantaneously.

5. User selects an item card to open the detail view.
6. User clicks "**Contact Seller**", types an inquiry message, and submits.
7. System generates a direct messaging thread and dispatches a notification to the seller.

6. Project Scope Document

6.1 Executive Summary

CampusConnect is a centralized, web-based platform tailored exclusively for university ecosystems. The system bridges the gap between student marketplace needs, micro-entrepreneurship, peer services, event discovery, and early career opportunities.

6.2 System Boundaries & In-Scope Features

- **Authentication & Profiles:** Single Sign-On (SSO) via institutional email (that gotten from the school), role management, and customizable profiles.
- **Marketplace Module:** Listing creation, categorized browsing, direct buyer-seller messaging, and item management (Buy/Sell/Trade).
- **Service Exchange Module:** Freelance/peer service directory, portfolio uploads, and service request inquiries.
- **Career & Internship Portal:** Job/internship postings by verified employers/alumni, resume attachment, and application tracking.
- **Campus Events Hub:** Calendar-based event discovery, category tagging, and RSVP tracking.
- **Communication Suite:** Real-time, peer-to-peer messaging system with automated safety warnings.
- **Search & Discovery Engine:** Dynamic keyword search with multi-faceted filtering (price, category, proximity/campus, date).

6.3 Out-of-Scope

- Direct escrow(neutral third party that holds money or objects until specific conditions are met) or integrated payment gateway processing (transactions occur offline or via peer-to-peer payment apps).
- In-app video/audio calling.

- Native mobile applications (focuses on a responsive Progressive Web App - PWA).

6.4 Constraints & Assumptions

- **Constraints:** Launch must precede the start of the upcoming academic fall semester; system design must comply with relevant data privacy regulations (like that of GDPR(General Data Protection Regulation)).
- **Assumptions:** Users have access to valid university email addresses; event hosts provide accurate scheduling information.

7. Requirement Traceability Matrix (RTM)

The matrix below maps user needs, functional requirements, use cases, and verification methods to ensure full lifecycle coverage.

Requirement ID	Business Objective / User Story	Functional Requirement	Use Case Reference ID	Test Verification Method
REQ-01	US-01, US-02	FR-MKT-01	UC-01	Integration Testing, UI Automated Tests
REQ-02	US-06	FR-MSG-01	UC-02	End-to-End Messaging Test
REQ-03	US-01, US-04	FR-SCH-01	UC-02	Search Index Performance Test
REQ-04	US-03	FR-SRV-01	UC-01 (Variant)	Functional System Test
REQ-05	US-04	FR-JOB-01	UC-01 (Variant)	User Acceptance Testing (UAT)
REQ-06	System Security	FR-AUTH-01, FR-AUTH-02	All Use Cases	Security / Penetration Test